



Reg. No:

--	--	--	--	--	--	--	--	--	--	--

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2024

B.E – COMPUTER SCIENCE AND ENGINEERING

20CS2E11 - BIG DATA TECHNOLOGIES

Answer ALL questions

Duration: 2 Hrs 30 min

Maximum: 70 Marks

PART - A

(5x 1 = 5 Marks)

1. **CO1 What does the term “big data” refer to?**
 - a) Any large dataset
 - b) Data that cannot be processed
 - c) Data that is too complex to analyze
 - d) Large and complex datasets that require specialized tools and techniques
2. **CO2 Hadoop is a framework that works with a variety of related tools. Common cohorts include _____**
 - a) MapReduce, Hive and Hbase
 - b) MapReduce, MySQL and Google Apps
 - c) MapReduce, Hummer and Iguana
 - d) MapReduce, Heron and Trumpet
3. **CO2 On which of the following platforms does Hadoop run?**
 - a) Debian
 - b) Cross-platform
 - c) Bare metal
 - d) Unix-like
4. **CO2 According to analysts, for what can traditional IT systems provide a foundation when they're integrated with big data technologies like Hadoop?**
 - a) Big data management and data mining
 - b) Data warehousing and business intelligence
 - c) Management of Hadoop clusters
 - d) Collecting and storing unstructured data
5. **CO3 _____ can best be described as a programming model used to develop Hadoop-based applications that can process massive amounts of data.**
 - a) MapReduce
 - b) Mahout
 - c) Oozie
 - d) Avro

PART - B

(5x 3 = 15 Marks)

6. **CO1** Describe the significance of 5'V's in Big Data.
7. **CO1** List the challenges of Big data.
8. **CO1** Define CAP Theorem.
9. **CO3** Classify the terms Avro and Oozie.
10. **CO2** Why the Partitions are shuffled in Map Reduce?

PART - C

(5 x 10 = 50 Marks)

Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. **CO1** Illustrate the main characteristics of Big data architecture with a neat schematic diagram. (10)
12. **CO2** Classify the components of Hadoop Framework. (10)
13. **CO1** Explain in detail about the challenges of conventional system. (10)
14. **CO2** Generalize how the data flow takes place in Mapreduce framework. (10)
15. **CO2** Explain the roles of the JobTracker and TaskTracker in the MapReduce function. (10)
16. **CO3** Illustrate the Apache Hadoop Core Components with architecture. (10)
17. **CO3** Summarize the following Bigdata eco system tools in detail. (10)
 - Hive
 - Pig
 - Flume
 - Sqoop
 - Zookeeper
